

Locker-Security Alarm

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Presented here is a simple locker security alarm that can be used to protect a locker from unauthorised access. The circuit is low-cost and forms a fool-proof alarm that receives its control signal from a standard reed switch. The circuit works off a 12V DC power supply.

Circuit and working

Fig. 1 shows the circuit diagram of the locker-security alarm. The circuit is built around quad NOR gate IC CD4001 (IC1), reed switch RS1 and transistor BC547 (T1). Reed switch RS1, together with a permanent magnet, forms the sensing unit as shown in Fig. 1.

The magnet is fixed on the door of the locker and reed switch RS1 on the door-frame such that when the door closes, both of them come close to each other. Now, when switch S1 is turned

on, 12V supply is extended to the circuit. Normally, when the locker is closed, the reed switch is also closed due to the presence of bar magnet, and hence

driver transistor T1 activates to drive the alarm. A hooter at CON3 or a buzzer at CON2 can be connected for the audio alarm. A separate power supply in case of hooter needs to be connected at CON4. Push-to-on-type switch S2 is used to reset the bistable circuit.

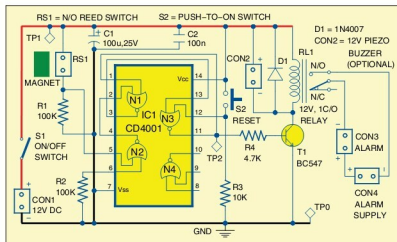


Fig. 1: Circuit diagram of the locker-security alarm

Working of the circuit is straightforward. When someone opens the locker door, reed switch RS1 opens and an alarm is raised. The alarm remains enabled even if the door is closed. The

alarm is turned off once switch S2 is pressed.

Construction and testing

An actual-size, single-side PCB for the locker-security alarm is shown in Fig. 2 and its component layout in Fig. 3.

After assembling the circuit on PCB, enclose it in a suitable plastic case. Fix the magnet on the door and the reed switch on the door-frame. Connect the reed switch through wires to RS1 on the PCB. Connect a buzzer at CON2 or a hooter at CON3. The power supply for the hooter is connected at CON4. Finally, connect the power supply for the circuit at CON1 and the circuit is ready to use. Take care that all the connectors and switches stick

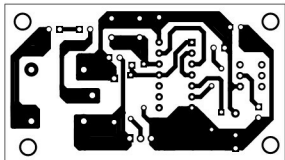


Fig. 2: An actual-size, single-side PCB of the locker-security alarm

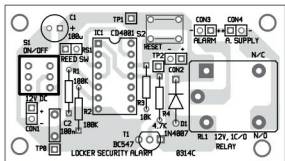


Fig. 3: Component layout for the PCB

out of the plastic case.

To test the circuit for proper functioning, verify various voltage levels as indicated in the test point table. ●

The author is an M.Tech from NIT, Raipur (Chhattisgarh) and deeply interested in designing electronics circuits

Test Points

Test point	Details
TP0	GND
TP1	12V (when S1 is on)
TP2	High when the locker door is opened

PARTS LIST

Semiconductors:

- IC1 - CD4001 quad NOR gate
- D1 - 1N4007 rectifier diode
- T1 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2 - 100-kilo-ohm
- R3 - 10-kilo-ohm
- R4 - 4.7-kilo-ohm

Capacitors:

- C1 - 100 μ F, 25V electrolytic
- C2 - 100nF ceramic disk

Miscellaneous:

- S1 - On/off switch
- S2 - Push-to-on switch
- RS1 - N/O reed switch
- CON1-CON4 - 2-pin connector
- RL1 - 12V, 1C/O relay
- Permanent magnet
- 14-pin IC base
- 12V battery or 12V, 500mA regulated supply